

AI Agents Introduction

Beyond Chatbots — AI That Plans, Acts, and Adapts

The Agent Loop | Claude & OpenAI Agents | Use Cases | Safety

What You Will Learn

- What an AI agent is and exactly how it differs from a chatbot or simple automation
- The agent loop: how perceive-plan-act-observe cycles let AI complete real tasks
- Real agent platforms compared: Claude tool use, OpenAI Operator, AutoGPT, CrewAI
- 6 practical agent use cases for engineers, plus the safety rules that keep you in control

The One-Line Definition

A chatbot only talks. An AI agent has tools, memory, and a goal —
so it can actually go and DO the task, then check its own work.

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AI-012: AI Agents Introduction

Up to now in this series, you have used AI the way most people do: you open a chat window, type a request, read the reply, and decide what to do with it yourself. That is a **chatbot interaction** — useful, but limited to a single back-and-forth. An **AI agent** is a different kind of system. Give it a goal, and it will plan the steps needed to reach that goal, use tools to carry out each step — searching the web, running code, reading files, calling other software — check whether the result actually moved it closer to the goal, and keep going until the task is done or it needs your input.

The shift from chatbot to agent is the single biggest change happening in AI right now. It is the difference between asking someone for directions and handing them the keys and saying "drive us there." This lesson gives you a clear, practical understanding of what agents are, how they work under the hood, which real platforms you can use today, and — just as importantly — how to keep yourself in control while using them.

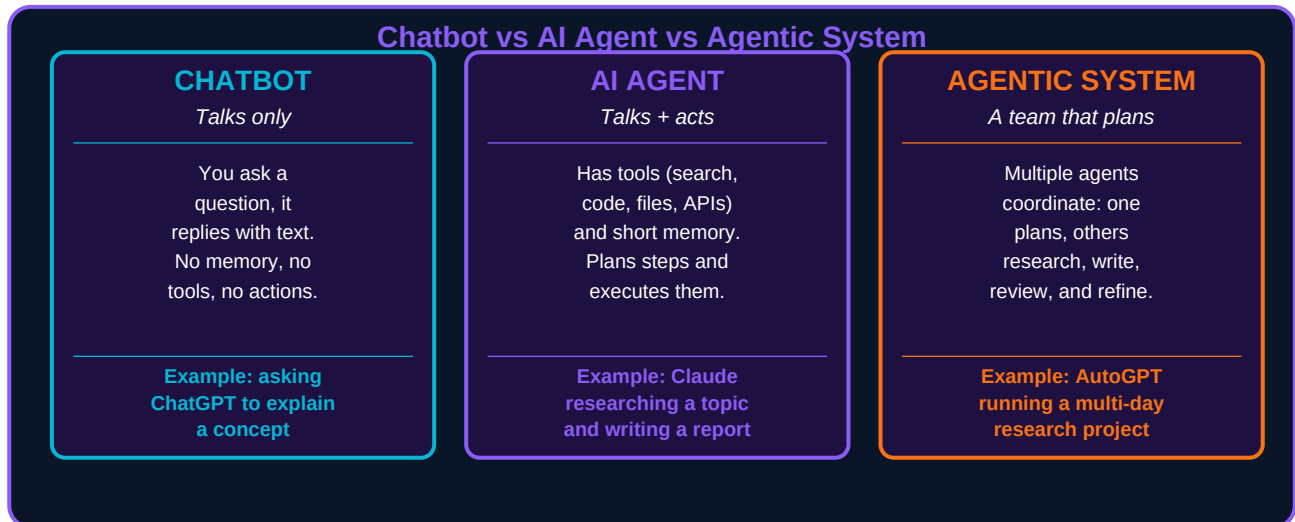
Learning Objectives

- Define an AI agent and explain precisely how it differs from a chatbot and from simple automation
- Describe the agent loop (perceive → plan → act → observe) and why it lets AI complete multi-step tasks
- Compare real agent platforms — Claude tool use / Computer Use, OpenAI Operator, AutoGPT, CrewAI
- Identify 6 practical ways engineers can use AI agents in their daily work
- Apply safety rules and human checkpoints so agents stay helpful tools, not unsupervised risks

1. Chatbot vs AI Agent vs Agentic System

System	What It Has	What It Can Do	Limitation
Chatbot	Language model only — no tools, no persistent memory	Answers questions, writes text, explains ideas — in a single reply	Cannot take any action. Cannot verify its own answer. Forgets everything after the chat ends.
AI Agent	Language model + tools (search, code execution, files, APIs) + short-term memory	Plans a sequence of steps, executes them with real tools, checks results, and adjusts	Works on one task at a time. Still needs a human to set the goal and review the final output.
Agentic System	Multiple specialised agents + a coordinator/planner that assigns and reviews work	Runs entire multi-day projects: research, draft, review, revise — with agents checking each other	Early-stage technology. Expensive to run. Errors in one agent can cascade through the whole system.

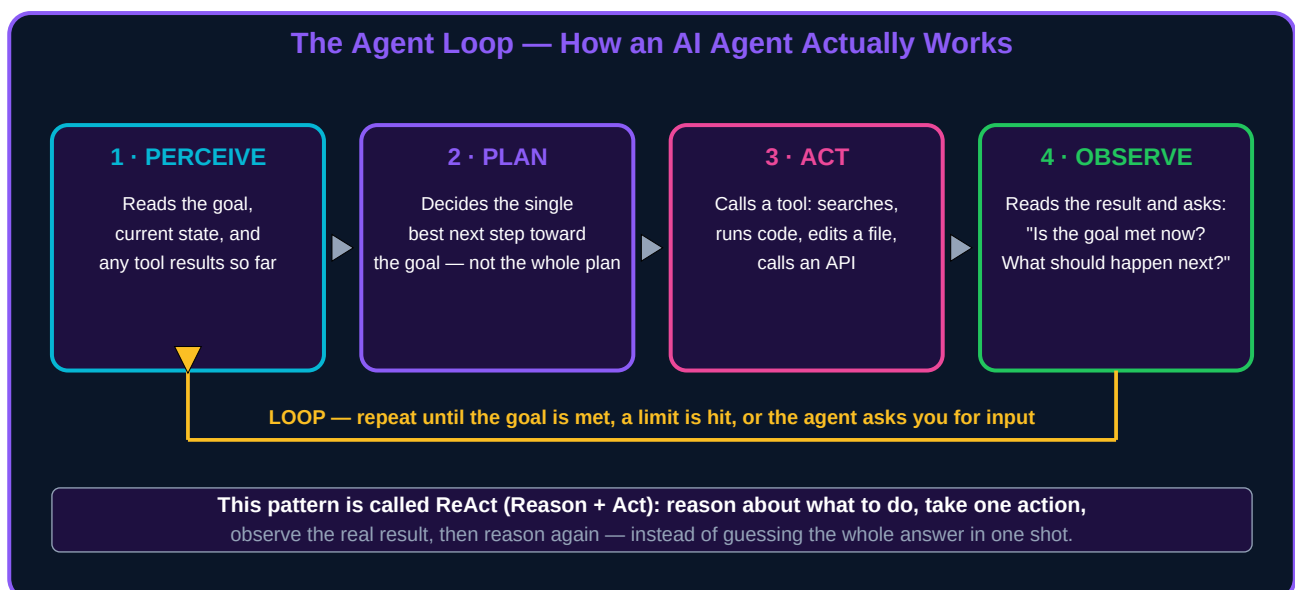
2. The Agent Loop — How Agents Actually Work



Every AI agent — no matter how advanced — runs on the same underlying cycle. Understanding this cycle is the key to understanding what agents can and cannot reliably do:

- **Perceive:** The agent reads the goal you gave it, the current state of the task, and the results of any tools it has already used.
- **Plan:** Based on what it perceives, it decides the single best next step toward the goal — not the whole plan at once, just the next move.
- **Act:** It actually carries out that step using a tool — searching the web, running a script, editing a file, calling an API, clicking a button.
- **Observe:** It reads the result of that action and asks itself: "Did this get me closer to the goal? Is the goal now complete? What should I do next?"

3. Visualising the Loop — Perceive, Plan, Act, Observe



This pattern has a name in AI research: **ReAct** (Reason + Act). The agent reasons about what to do, takes one action, observes what happened, and reasons again — rather than trying to plan an entire 20-step task perfectly in advance (which AI is bad at) or blindly executing one giant instruction with no checking (which leads to compounding errors).

Why the Loop Matters More Than the Model

Without a loop (single prompt)	With a loop (agent)
"Write a script that calculates beam deflection, run it, and tell me if it works." — the AI guesses at the whole answer in one shot, including code it has never run.	The agent writes the script, actually RUNS it, reads the real error message if one appears, fixes the specific line that failed, and re-runs — repeating until it works.
If the first guess is wrong, you get a broken result with no indication of where it failed.	Each step is checked against reality before the next one begins — closer to how a careful engineer actually works.
Best for: quick drafts where you will review and fix everything yourself.	Best for: tasks where "try it, check it, fix it" is exactly the workflow you would use yourself.

4. What Makes an Agent "Agentic" — Three Ingredients

- **Tools:** Without tools, an agent is just a chatbot. Tools are what let it search the live web, execute code in a sandbox, read and write files, send messages, or call external services through APIs.
- **Memory:** The agent needs to remember what it has already tried within the current task — otherwise it would repeat the same failed step forever. Most agents keep a working memory of recent steps and results.
- **Autonomy with limits:** The agent decides HOW to reach the goal — which tools to use, in what order, how many attempts to make — within boundaries that you, the human, define in advance.

Notice that none of these three ingredients require the AI model itself to be smarter than the chatbot you already use. The same underlying model — Claude, GPT, Gemini — becomes an agent simply by being wrapped in a system that gives it tools, a working memory, and a clearly bounded goal. This is why the platforms below all feel different in capability even though several of them are powered by similar underlying models: the difference is in how much "agency" the surrounding system grants them.

5. Real Agent Platforms You Can Use Today

Real AI Agent Platforms — What They Can Actually Do

CLAUDE

Computer Use / Tool Use
anthropic.com

Browses, clicks, types,
runs code, reads and
writes files. Built into
Claude Desktop & API.

BEST FOR

Research, coding,
documents, analysis

OPENAI OPERATOR

ChatGPT Agent
openai.com

Browses the web, fills
forms, books and orders
things, runs code in a
sandboxed browser.

BEST FOR

Web tasks, shopping,
browser automation

AUTOGPT / CREWAI

LangChain Agents
github.com

Open-source multi-agent
frameworks. You define
agent roles and they
coordinate a project.

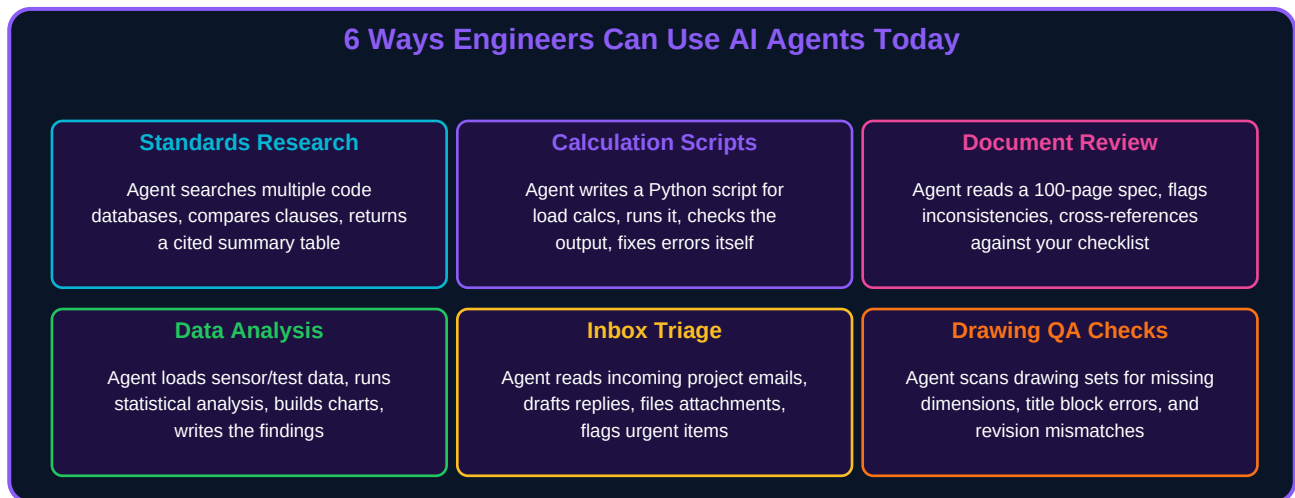
BEST FOR

Developers building
custom agent systems

Platform	How You Access It	Strongest At	Watch Out For
Claude (tool use / Computer Use)	Claude.ai, Claude Desktop, Anthropic API	Research, coding, document analysis, multi-step reasoning tasks	Computer Use (controlling your screen) is powerful — always supervise it directly while it runs
OpenAI Operator / ChatGPT Agent	ChatGPT Plus/Pro, OpenAI API	Browsing the live web, filling forms, completing online tasks end-to-end	Can take real-world actions like purchases — always set spending and action limits first
AutoGPT / CrewAI / LangChain Agents	Open source — free to run, requires setup	Building fully custom multi-agent systems for specific business workflows	Requires technical setup. Can loop or run up unexpected API costs if not capped
Microsoft Copilot Agents	Microsoft 365, Copilot Studio	Office-integrated tasks: emails, documents, spreadsheets, Teams workflows	Best inside the Microsoft ecosystem — less flexible for tools outside it

6. AI Agent Use Cases for Engineers

You do not need to be a programmer to start using agents productively. These six use cases use agent features that are already built into tools you may already have access to — Claude with tool use, ChatGPT with Code Interpreter and browsing, or Copilot in Microsoft 365.



Picking your first use case: start with whichever task on this page currently eats the most of your week and has the lowest cost if the agent gets something wrong on the first try. Standards research and document review are good starting points for most engineers — the agent produces a draft summary or flagged list, and you stay firmly in the loop as the final reviewer who checks it against the source before anything goes into a real deliverable.

- **This week:** pick ONE use case above, give the agent a real (but low-stakes) task from your current workload, and compare its draft against what you would have produced manually.
- **Track the time saved:** note how long the agent took versus your usual time, and how much of its output you kept versus rewrote — this tells you where agents are already strong enough to trust, and where you still need to do the work yourself.

7. Building Your First Agent Workflow — A Walkthrough

Here is exactly how to run your first real agent task using Claude with tool use (or ChatGPT with browsing + Code Interpreter enabled). This example has the agent research a topic, verify its sources, and produce a structured summary — a task that would normally take you 45–60 minutes.

Step	What You Do	What the Agent Does
1. Set the goal	Type: "Research the current IEC and IEEE standards for low-voltage cable sizing. Compare their ampacity derating methods and summarise the key differences in a table with sources."	Reads the goal, breaks it into sub-tasks: find IEC standard info, find IEEE standard info, compare, format as table
2. Agent searches	(No action needed — observe)	Uses its search tool to find official standard documents and reputable engineering references for both IEC and IEEE
3. Agent extracts	(No action needed — observe)	Reads the search results, extracts the specific derating method and figures from each standard
4. Agent verifies	(No action needed — observe)	Cross-checks the figures against a second source for each standard before including them — reducing the risk of an isolated error
5. Agent formats	(No action needed — observe)	Builds a comparison table: Standard Method Key Formula Typical Derating Factor Source link
6. You review	Read the table. Click through to the cited sources. Confirm the figures against your own copy of the standard before using them in real work.	Waits for your follow-up — refine, expand, or move to the next task

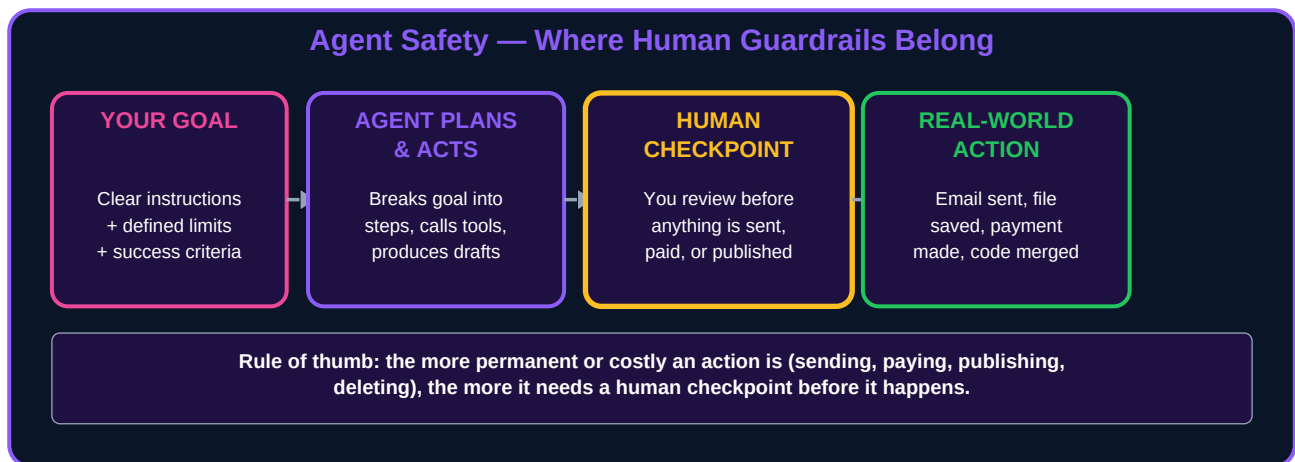
8. Agent Safety — Staying in Control

An agent that can take real actions is more useful — and carries more responsibility — than a chatbot that only talks. The goal is not to avoid agents, but to put the right checkpoints in the right places, just as you would with any new team member learning your processes.

Before you hand any real task to an agent, ask yourself three questions:

- What is the worst thing that could happen if the agent gets this completely wrong, and would I notice before it caused real damage?
- Which step in this task is the one I would most want to check myself before it goes any further — and have I told the agent to stop there?

- If the agent ran this task ten times, would nine good results and one bad one be an acceptable trade-off, or does even one bad result matter too much here?



In practice, the "Human Checkpoint" stage above is where most of your real control lives. It is the single point where you decide whether a draft becomes a sent email, a calculation becomes a submitted design value, or a summary becomes the version your team relies on. Treat it the same way you would treat reviewing a junior engineer's first submission on a task — confirm it, correct it where needed, and only then let it move forward.

- **Set the checkpoint before you start, not after:** decide in advance which actions need your sign-off (e.g. "never send an email without showing me the draft first") rather than discovering the gap once something has already gone out.
- **The checkpoint gets lighter as trust grows:** early on, review everything in detail. Once an agent has proven reliable on a specific, narrow task, you can review its output more lightly — but keep spot-checking, since agents can drift in quality as tasks change.

None of this is unique to AI — it is the same judgment you already use when delegating to a colleague, a contractor, or a new hire. The five rules on the next page turn that judgment into a short checklist you can apply to any agent task, regardless of which platform or model is doing the work.

A simple way to picture the difference a checkpoint makes: imagine asking an agent to "email the client an updated project schedule." Without a checkpoint, a single misread date or wrong attachment goes straight into their inbox — and you find out only when they reply confused. With a checkpoint, the same mistake appears as a draft on your screen, where it costs you ten seconds to catch and fix instead of an awkward phone call to undo.

9. The Five Safety Rules for Working With Agents

Rule	Why It Matters	How To Apply It
1. Define clear limits up front	Agents will do exactly what you ask — including things you did not consider	State explicitly: "Do not send anything. Do not spend money. Stop and ask me before X."
2. Review before anything goes out	An agent's output looks polished, which can create false confidence	Treat every agent output as a draft from a junior colleague — read it fully before it is used
3. Verify facts and figures independently	Agents can confidently present incorrect information as fact	Cross-check any number, standard, or citation against the original source before relying on it
4. Start with low-stakes tasks	You learn an agent's strengths and blind spots fastest on tasks where mistakes cost little	Practice with research, drafts, and analysis before trusting agents with anything client-facing
5. Set hard limits on autonomy	Long unsupervised runs are where small errors compound into large ones	Cap the number of steps or the time an agent can run before it must check back in with you

10. Practice Exercises

#	Exercise	Instructions
1	Run your first agent research task	Open Claude or ChatGPT with browsing/tool use enabled. Give it the research task from Section 7 (or a topic from your own field). Watch HOW it works — note each step it takes, not just the final answer. Write down 3 things it did that surprised you.
2	Test the loop yourself	Ask an agent to "write a Python script that converts Celsius to Fahrenheit, run it, and confirm the output is correct for 0°C, 100°C, and -40°C." Observe how it handles an error if one occurs — does it notice, explain, and fix it on its own?
3	Define your own safety limits	Before your next agent session, write down — in plain language — exactly what you will and will not allow the agent to do without checking with you first. Paste those limits into your prompt as explicit instructions.
4	Compare chatbot vs agent on the same task	Pick one task (e.g. "find the three most relevant building codes for a rooftop solar installation"). First ask a plain chatbot. Then ask an agent with search enabled. Compare the depth, accuracy, and verifiability of each answer.

5	Map an agent workflow for your own work	Identify ONE multi-step task you do regularly (research + summary, data check + report, inbox triage + draft replies). Write it out as: Goal → Tools needed → Checkpoints → Final review. This is the blueprint for your first real agent workflow.
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Lesson Summary

Concept	Key Rule
Chatbot vs Agent	A chatbot only replies. An agent has tools, memory, and a goal — so it can plan, act, and check its own results.
The Agent Loop (ReAct)	Perceive → Plan → Act → Observe — repeated until the goal is met. This is what lets agents handle multi-step tasks reliably.
Three ingredients of "agentic"	Tools (to act) + Memory (to track progress) + Bounded autonomy (to decide HOW within limits YOU set).
Best agent platforms today	Claude (tool use/Computer Use) for research & coding, OpenAI Operator for web tasks, AutoGPT/CrewAI for custom multi-agent systems.
Where engineers benefit most	Standards research, calculation scripts, document review, data analysis, inbox triage, drawing QA — always with human review.
The golden safety rule	The more permanent or costly an action (sending, paying, publishing, deleting), the more it needs a human checkpoint first.

What Is Next — AI-013

AI-013: AI for Business Automation — Taking the agent concepts from this lesson and applying them at the business-systems level: customer support agents, lead qualification pipelines, automated reporting systems, and how to design an "agent team" that runs a real business process end to end with proper human oversight built in.

Resource	Details
AI PDF Series	AI-001 through AI-030 — ayetechub.com/pdfs.html
AI Tools Directory	Claude, ChatGPT, agent frameworks reviewed — ayetechub.com/ai-tools.html
Community	Telegram: t.me/ayetechub